

1. Proticity of the acid (x) $\left\{ \begin{array}{c} H_xA (aq) \longrightarrow xH^+ (aq) + A^{x-} (aq) \\ 1 \text{ mol} \qquad \qquad x \text{ mol} \qquad \qquad 1 \text{ mol} \end{array} \right\}$
2. mass of the solid acid ($m_{a,i}$) $i \Rightarrow n^{\text{th}} \text{ trial}$
3. molarity of NaOH (M_{NaOH})
4. Volume of NaOH used for titration ($V_{NaOH,i} = V_2 - V_1$)
titrated vol.
5. At the equivalence point
 $n_{NaOH,i} = n_{H^+,i}$
6. $n_{NaOH,i} = M_{NaOH} \times V_{NaOH,i} \Rightarrow \text{from (5)} \quad n_{H^+,i} = M_{NaOH} \times V_{NaOH,i}$
7. It follows from (1)

$$n_{acid,i} = \frac{n_{H^+,i}}{x} \quad \text{Also,} \quad n_{acid,i} = \frac{m_{acid,i} [g]}{M_{acid} [g \text{ mol}^{-1}]}$$

$$8. M_{acid,i} [g \text{ mol}^{-1}] = m_{acid,i} / n_{acid,i}$$

from (7)

$$M_{acid,i} = \frac{m_{acid,i} \times x}{n_{H^+,i}}$$

from (6)

$$M_{acid,i} = \frac{m_{acid,i} \times x}{M_{NaOH} \times V_{NaOH,i}}$$

$\therefore$ nonreducible form

$$M_{acid,i} = \frac{x \times m_{acid,i}}{M_{NaOH} \times V_{NaOH,i}}$$

Average ($\bar{M}$)

$$\bar{M}_{acid} = \frac{x}{M_{NaOH}} \cdot \sum_{i=1}^n \left(\frac{m_{acid,i}}{V_{NaOH,i}} \right)$$